Erol Cetinok

Yale MechE + CS | AI/ML Engineering Intern @ Ceramic.ai

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Experience

Ceramic.ai — Sep 2025 – Present
AI/ML Engineering Intern — *San Francisco, CA*

- Web-scale search that's 100× cheaper, 10× faster, accurate, and fully independent.
- Contribute to Ceramic's search infrastructure, focused on evaluation systems for search quality.

FRC Team 751 — Aug 2022 – May 2026
Component Lead — *Portola Valley, CA*

- Consistently decreased robot cycle time through my design of intakes, end-effectors, an elevator, a hopper mechanism, and a driver station across four seasons.
- Extensive design-for-manufacture experience with CAD software, the milling machine, lathe, and CNC.

University of Florida — Flux Lab — Jun – Aug 2025
Research Intern — *Gainesville, FL*

- Conducted research on heat transfer and transport phenomena for PCM-integrated building envelopes.
- Presented a "Techno-Economic Analysis of Conductor Selection for Thermal Switch and PCM Systems" at the MIT Undergraduate Research Conference.

Projects

Full-Stack Quadruped Robot — Jan 2026 – Present
Engineered a 12-DoF quadruped spanning CAD, Arduino/C++ firmware, and a Python control stack, with analytic inverse kinematics and a gait generator, validated in MuJoCo via a swappable sim/hardware backend.

RoboForAll — Open-Source Classroom Robotics Kit — Jan – May 2026
Built a low-cost, replicable educational robot (ESP32, 3D-printed chassis) with a web platform (roboforall.org) featuring a Blockly studio, classroom curriculum, and in-browser simulator.

Bio-Inspired Wall-Climbing Robot — Oct 2025 – Jan 2026
Designed and built a microspine wall-climbing robot inspired by JPL's compliant-roller climbers, with a 3D-printed chassis, roller-mounted compliant mechanisms, and RC-car electronics.

Custom Hardtail Conversion Linkage — Nov 2024 – Feb 2025
Engineered a custom aluminum linkage converting a full-suspension mountain bike into a hardtail; FEA-optimized, CNC-machined, and anodized, now in regular use.

Education

Yale University — 2026 – 2030
B.S. Mechanical Engineering; B.A. Computer Science — *New Haven, CT*

Woodside Priory School — 2019 – 2026
High School Diploma — *Portola Valley, CA*

Volunteering: Section Leader, Stanford Code in Place — teaching introductory Python (Apr 2025 – Present).

Technical Skills

Languages & Web: Python, C/C++, Java, JavaScript, TypeScript, Next.js, React

ML & Robotics: PyTorch, ROS, MuJoCo, Arduino, MATLAB

CAD & Manufacturing: SOLIDWORKS, Fusion 360, AutoCAD, FEA, CNC, milling, lathe, MIG welding, 3D printing

Honors & Awards

Computer Science Department Award (2026) • History & Social Sciences Department Award (2025) • Model UN: Best Delegate (SVMUN 2025), Best Research (SVMUN 2026) • Bay to Breakers 15K: 2nd in age group, 31st of 3,658.